

Separable commutative algebras in stable module categories

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ABSTRACT. Let k be a separably closed field of characteristic $p > 0$. We prove that every separable commutative algebra in the stable module category of a finite p -group is the function algebra of a finite p -group set. After free orbits are removed, the function-algebra construction is an equivalence of categories. For general finite groups, we give a negative answer to Balmer's question: in characteristic three, a 28-dimensional orthogonal summand of the permutation module on three-element subsets of a seven-element set yields a separable algebra $k \oplus L$ in $\text{stmod}(k\mathfrak{A}_7)$ whose underlying stable object is not a permutation module. The counterexample is proved by explicit module calculations. The classification uses a relative norm on invariant vectors, separable algebras over torus Tate cochains, and descent through normal maximal subgroups. In characteristic two, power operations force a rational point in the support; a commutative residue algebra at that point supplies the local freeness needed for the torus classification.

CONTENTS

1. Introduction	1
The two proof routes	3
Conventions	3
2. Stable modules and projective factorizations	4
3. A counterexample for $\mathfrak{A}_7$	6
3.1. The orthogonal summand	6
3.2. A separable quadratic algebra	8
3.3. Fixed points of permutation modules	9
3.4. Completion of the counterexample	11
4. Graph idempotents and splitting towers	11
5. A relative norm on invariant vectors	13
6. Spectral inputs and torus descent	15
7. Characteristic two: a rational point in the support	17
8. Characteristic two: residue algebras and full support	19
9. Classification over the torus Tate ring	20
10. Elementary abelian groups	21
11. Standard algebras and group actions	22
12. Gluing over normal maximal subgroups	24
References	26

1. INTRODUCTION

For a finite group G and a field k of characteristic $p > 0$, write $\text{stmod}(kG)$ for the category of finite-dimensional left kG -modules modulo maps factoring through projective modules. Tensor

products are over k , with diagonal G -action. A commutative algebra object in this category is called *separable* if its multiplication admits a bimodule section.

Every finite G -set X gives such an algebra. Its underlying module is the permutation module kX , with basis $(e_x)_{x \in X}$ and function-algebra structure

$$e_x e_y = \delta_{x,y} e_x, \quad 1_{kX} = \sum_{x \in X} e_x. \quad (1.1)$$

The map $e_x \mapsto e_x \otimes e_x$ splits multiplication as a bimodule map. Over a separably closed field, every finite-dimensional separable commutative algebra with a G -action has this form in the ordinary module category: its primitive idempotents form a finite G -set. Balmer asked whether the same description holds after passing to the stable category [Bal15, Question 4.7].

Question (Balmer). Let k be separably closed. Is every separable commutative algebra $A \in \text{stmod}(kG)$ isomorphic, as an algebra, to kX for some finite G -set X ?

We call algebras of the form kX *standard*. Pullback of functions makes $X \mapsto kX$ contravariant. A free orbit gives a projective module, so its function algebra is zero in the stable category. For a finite p -group P , let Fin_P^{nf} be the full category of finite P -sets with no free P -orbit, including the empty set.

Theorem 1.1 (Finite p -groups). *Let k be separably closed of characteristic $p > 0$, and let P be a finite p -group. The function-algebra functor is an equivalence*

$$(\text{Fin}_P^{\text{nf}})^{\text{op}} \xrightarrow{\sim} \text{CAlg}^{\text{sep}}(\text{stmod}(kP)). \quad (1.2)$$

In particular, every separable commutative algebra has a unique decomposition, up to reordering and conjugacy of subgroups,

$$A \simeq \prod_{[1 \neq H \leq P]} k(P/H)^{\times m_H}, \quad m_H \in \mathbb{Z}_{\geq 0}. \quad (1.3)$$

The empty product is the zero algebra.

For $P = 1$, both categories in (1.2) have only one object up to unique isomorphism: the empty set on the left and the zero algebra on the right. For general finite groups, the answer to Balmer's question is negative. The next theorem gives a counterexample already at the level of underlying stable objects. Write $\mathfrak{A}_7$ for the alternating group on seven letters.

Theorem 1.2. *Let k be a field of characteristic 3, and let*

$$\Omega = \{T \subseteq \{1, \dots, 7\} : |T| = 3\}.$$

The $k\mathfrak{A}_7$ -submodule

$$L = \left\{ (x_T)_{T \in \Omega} \in k\Omega : \sum_{T \in \Omega} x_T = 0, \quad \sum_{T \ni i} x_T = 0 \text{ for } 1 \leq i \leq 7 \right\} \quad (1.4)$$

has dimension 28, and the symmetric bilinear form

$$b(x, y) = \sum_{T \in \Omega} x_T y_T \quad (1.5)$$

is nondegenerate and $\mathfrak{A}_7$ -invariant on L . The module $A = k \oplus L$, with unit $(1, 0)$ and multiplication

$$(a, x)(a', y) = (aa' + b(x, y), ay + a'x), \quad (1.6)$$

is a separable commutative algebra object in $\text{stmod}(k\mathfrak{A}_7)$. For every finite $\mathfrak{A}_7$ -set X , its underlying stable object satisfies

$$A \not\sim_{\text{st}} kX.$$

In particular, over $k = \overline{\mathbb{F}}_3$ this gives a negative answer to Balmer's question for general finite groups.

The counterexample works over every field of characteristic three; taking $k = \overline{\mathbb{F}}_3$ places it in the setting of Balmer's question. It does not conflict with Theorem 1.1, since $\mathfrak{A}_7$ is not a 3-group. In particular, the distinction between the two results is one of groups, rather than a change in the meaning of separability.

Balmer and Carlson proved the classification for cyclic p -groups [BC18, Theorem 4.1]. Naumann and Pol obtained a positive answer for algebras of finite splitting-tower degree in finite groups of p -rank one [NP24, Theorem 14.14]. We prove the degree bound needed here independently in Proposition 4.4. We use their regular even-periodic classification only where 2 is invertible [NP24, Theorem 10.7]; the characteristic-two argument below supplies the required torus result separately. Separable lifting and algebra descent are provided by Ramzi [Ram23, Theorem B] and by the formulation in Naumann–Pol–Ramzi [NPR26, Theorem 2.1 and Proposition 2.3]. The conclusions of Theorems 1.1 and 1.2 are proved here from these inputs and the module calculations below.

The two proof routes. The counterexample requires two assertions about its orthogonal module: its invariant form identifies $L \otimes L$ with k modulo a specified projective kernel, and L restricts to a trivial module plus free modules on a Sylow 3-subgroup. The first assertion makes the explicit quadratic multiplication on $k \oplus L$ associative and separable in the stable category. The second constrains every putative permutation representative. A connected graph of 3-cycle subgroups then forces that representative to be stably $k^{\oplus 2}$, whereas maps from the unit distinguish $k \oplus L$ from $k^{\oplus 2}$. This proof is independent of the classification for p -groups.

For Theorem 1.1, the first reduction concerns a connected algebra for an elementary abelian group. Graph idempotents give a nontrivial kernel for its stable group action. If the entire group acts stably trivially, torus descent lifts the algebra to torus Tate cochains, where we prove it is a product of units. Otherwise, in the inductive argument, a core representative restricts to a free module on a complement of the stable kernel. A relative norm turns its invariant vectors into a nonzero connected separable algebra for the smaller kernel group. Induction reduces that invariant module to one dimension, and a transfer calculation then makes the kernel action actually trivial. At that point the algebra is an ordinary separable function algebra.

The torus classification has a separate issue in characteristic two: the commutative residue algebra used in the local argument must be constructed, not merely postulated as a module quotient. Power operations first force a point defined over $\mathbb{F}_2$ in the support. An integral change of torus coordinates supplies an actual commutative residue algebra there. Local freeness gives full support, and the last nonzero splitting-tower layer becomes a descendable algebra that splits the original algebra. Only then do we apply the classification of finite covers of regular even-periodic rings.

Finally, for a non-elementary-abelian p -group, all maximal subgroups are normal and their nonempty intersections contain the nontrivial Frattini subgroup. Local standard algebras therefore determine finite sets with full group actions and compatible restrictions. We glue these sets one global orbit type at a time and apply subgroup descent. Full faithfulness of the function-algebra functor identifies the resulting objects and morphisms, completing (1.2).

Conventions. All modules in the algebraic parts are finite-dimensional left modules. We use $\cong$ for ordinary module isomorphisms and $\simeq_{\text{st}}$ for stable module isomorphisms; $\simeq$ between algebras denotes an isomorphism in the specified algebra category. The tensor symmetry is the usual interchange of factors, with no sign attached to module dimensions. For modules U, V , put

$$\underline{\text{Hom}}_{kG}(U, V) = \text{Hom}_{kG}(U, V) / \mathcal{P}_G(U, V),$$

where $\mathcal{P}_G(U, V)$ consists of maps factoring through finite-dimensional projectives. Such maps form an additive two-sided ideal. By Lemma 2.1, this ideal is also preserved by tensoring.

For clarity, a separable commutative algebra has a unit η , a commutative associative multiplication μ , and a map σ with

$$\mu\sigma = \text{id}_A, \quad \sigma\mu = (\mu \otimes 1)(1 \otimes \sigma) = (1 \otimes \mu)(\sigma \otimes 1). \quad (1.7)$$

The equalities are in the category under consideration. In a stable module category they need not hold for chosen ordinary representatives. The notation CAlg^{sep} includes all unital algebra maps between separable commutative algebras and includes the zero algebra. The field is assumed separably closed only when this is stated; in particular, no perfectness assumption on a separably closed field is implicit.

2. STABLE MODULES AND PROJECTIVE FACTORIZATIONS

Throughout, k is a field of characteristic $p > 0$.

Lemma 2.1 (Projectivity). *Let G be a finite group and $H \leq G$. Restriction and induction preserve finite-dimensional projective modules. If $[G : H]$ is invertible in k , a kG -module is projective whenever its restriction to H is projective. If F is projective and V is any finite-dimensional kG -module, then $F \otimes V$ is projective.*

Proof. The group algebra kG is free on both sides over kH , proving the first assertion. Put $m = [G : H]$. The induction counit

$$\epsilon : kG \otimes_{kH} U \downarrow_H \longrightarrow U, \quad g \otimes u \longmapsto gu$$

has, when m is invertible, the section

$$s(u) = m^{-1} \sum_{t \in \mathcal{T}} t \otimes t^{-1}u,$$

where $\mathcal{T}$ is a set of left coset representatives. Replacing t by th does not change its summand. If $gt = t'h$, then $gt \otimes t^{-1}u = t' \otimes t'^{-1}gu$, so s is G -linear. Thus U is a summand of the induction of its restriction.

Finally, the map

$$(kG \otimes V)_{\text{diagonal}} \longrightarrow (kG \otimes V)_{\text{first factor}}, \quad g \otimes v \longmapsto g \otimes g^{-1}v$$

is G -linear, with inverse $g \otimes v \mapsto g \otimes gv$. Its target is free of rank $\dim_k V$. The tensor assertion follows by taking direct summands. $\square$

Lemma 2.2 (Cancellation and stable summands). *Finite-dimensional modules over a finite-dimensional k -algebra have unique finite decompositions into indecomposable modules, up to isomorphism and permutation. If two modules are isomorphic modulo maps factoring through projectives, their nonprojective indecomposable summands agree as ordinary modules, with multiplicities.*

Proof. Existence follows by induction on dimension. For an endomorphism f of an indecomposable module U , the Fitting decomposition gives $U = \ker f^n \oplus \text{im } f^n$ for large n . Indeed, stabilization of kernels implies that the intersection is zero, and rank-nullity gives the sum. Thus f is invertible or nilpotent. The nonunits in $\text{End}(U)$ form an ideal: a nonunit a has $1 - a$ invertible, and if nonunits a, b had invertible sum, both $a(a + b)^{-1}$ and its complement would be nonunits, a contradiction. Multiplication by arbitrary elements preserves nonunits because a one-sided inverse in a finite-dimensional algebra is two-sided. Hence $\text{End}(U)$ is local.

In an additive category, compare two finite decompositions into objects with local endomorphism rings. For a summand U of one decomposition, the identity is a sum of composites $U \rightarrow V_j \rightarrow U$ through the other. One composite is invertible. Thus U is a retract of that V_j ; the resulting nonzero idempotent in the local ring $\text{End}(V_j)$ must be its identity. Hence $U \cong V_j$. Invertible block operations in the original isomorphism cancel this pair, and induction proves uniqueness and ordinary direct-sum cancellation.

For a nonprojective indecomposable U , the ideal I_U of maps through projectives is proper: $1_U \in I_U$ would make U projective. Therefore $\text{End}(U)/I_U$ is a nonzero local ring. The same uniqueness argument applies after passing to the quotient and omitting projective summands. A stable isomorphism between nonprojective indecomposables is ordinary: choose representatives

$f : U \rightarrow V$, $g : V \rightarrow U$ with $gf = 1 + t$, $t \in I_U$. This is a unit, so f is a split injection, and indecomposability of V makes f an isomorphism. $\square$

Lemma 2.3 (Core dimension). *Let G be any finite group. Every finite-dimensional kG -module has a decomposition $M = M_0 \oplus F$, with F projective and M_0 having no nonzero projective summand. The isomorphism type of M_0 is a stable invariant. Write $c(M) = \dim_k M_0$. Then*

$$c(U \oplus V) = c(U) + c(V), \quad c(U \otimes V) \leq c(U)c(V). \quad (2.1)$$

Every representative of a stable isomorphism between modules without projective summands is an ordinary module isomorphism.

Proof. The decomposition and its stable invariance follow from Lemma 2.2. The ideal of endomorphisms factoring through projectives lies in the radical of $\text{End}(M_0)$. To see this, decompose M_0 into indecomposables and pass to the semisimple quotient of its endomorphism ring, a product of matrix rings indexed by the distinct indecomposable types. Each entry from a projective factorization is zero there, since its ideal on an indecomposable nonprojective is proper in a local ring. If $f : M_0 \rightarrow N_0$ has stable inverse g , both gf and fg are identities plus radical elements and therefore are units. Hence f is an ordinary isomorphism.

Additivity of c follows from cancellation. In $U \otimes V$, every term involving a projective summand is projective by Lemma 2.1. Deleting any further projective summands from $U_0 \otimes V_0$ gives the tensor inequality. $\square$

We will use the idempotent completeness of $\text{stmod}(kG)$. It also follows from the preceding proof. On a core representative the projective-factorization ideal is contained in the radical of a finite-dimensional endomorphism ring and is therefore nilpotent. Idempotents lift modulo nilpotent ideals, so every stable idempotent is represented by an ordinary idempotent and splits as a module direct sum.

Lemma 2.4 (The group algebra of a p -group). *Let P be a finite p -group and put $N_P = \sum_{g \in P} g \in kP$. The algebra kP is local and self-injective, and*

$$\text{soc}(kP) = kN_P.$$

Every nonzero left ideal meets kN_P nontrivially. Every finite-dimensional projective kP -module is free.

Proof. The augmentation ideal J is nilpotent. Induct on $|P|$: for $P \neq 1$ choose a central element z of order p . In the quotient by $(z - 1)$, the claim is the induction hypothesis for $P/\langle z \rangle$. Thus $J^a \subseteq (z - 1)$ for some a , and $J^{ap} = 0$ since $(z - 1)$ is central and $(z - 1)^p = 0$. As $kP/J = k$, a finite geometric series inverts every element outside J , proving locality.

The coefficient of 1 in a product defines a nondegenerate associative symmetric pairing on kP . Give $(kP)^* = \text{Hom}_k(kP, k)$ the left action $(a \cdot f)(b) = f(ba)$. The map sending a to the functional $b \mapsto \text{coeff}_1(ba)$ identifies the left regular module with this dual, which is injective because $\text{Hom}_{kP}(-, (kP)^*) \cong \text{Hom}_k(-, k)$ is exact. A vector in the regular module killed by J has constant coefficients, giving the socle formula. For a nonzero left ideal I , its last nonzero term $J^a I$ lies in both I and the socle.

For a finite projective module U , lift a basis of U/JU to obtain a surjection $(kP)^d \rightarrow U$; its cokernel is zero by nilpotence of J . The surjection splits. Reducing the resulting direct sum modulo J makes its kernel have zero quotient by J , so that kernel is also zero. Thus $U \cong (kP)^d$. $\square$

Lemma 2.5 (Norms and transfers). *Let P be a finite p -group. If M has no projective summand, then*

$$N_P M = 0, \quad \underline{\text{Hom}}_{kP}(k, M) = M^P. \quad (2.2)$$

For finite-dimensional P -modules U, V , the maps factoring through projectives are exactly the images of

$$\mathrm{Tr}_P : \mathrm{Hom}_k(U, V) \longrightarrow \mathrm{Hom}_{kP}(U, V), \quad f \longmapsto \sum_{g \in P} gfg^{-1}. \quad (2.3)$$

Proof. If $N_P m \neq 0$, the map $kP \rightarrow M$, $a \mapsto am$, is injective: a nonzero kernel would meet kN_P . Self-injectivity splits this injection, contradicting the core assumption. A map $k \rightarrow kP$ lands in kN_P , so every map $k \rightarrow M$ through a projective is zero.

Each transfer factors through a free module, with action only on its first factor:

$$U \longrightarrow kP \otimes U \longrightarrow V, \quad u \longmapsto \sum_g g \otimes g^{-1}u, \quad g \otimes u \longmapsto gf(u).$$

Conversely, a P -map $U \rightarrow kP \otimes W$ has the form $u \mapsto \sum_g g \otimes a(g^{-1}u)$ for a k -linear $a : U \rightarrow W$, and a P -map from this free module to V has the form $g \otimes w \mapsto gb(w)$. Their composite is $\mathrm{Tr}_P(ba)$. Enlarge a projective factorization to a free one to obtain the claim. $\square$

Lemma 2.6 (Actual trivial action). *Let P be a finite p -group. Suppose $1 \neq H \leq P$ acts trivially on P -modules U, V . Then $\mathcal{P}_P(U, V) = 0$. Consequently, if a nontrivial subgroup acts trivially on a representative of a separable commutative algebra in $\mathrm{stmod}(kP)$, all its algebra and separability equations hold as ordinary module-map equations. When k is separably closed, the algebra is standard.*

Proof. A map $U \rightarrow (kP)^d$ lands in $((kP)^d)^H = (N_H kP)^d$. A map $(kP)^d \rightarrow V$ kills that space because N_H acts on V as $|H| = 0$. Apply this to all tensor powers in the algebra and bimodule-section identities. The resulting ordinary finite-dimensional separable commutative k -algebra is a product of finite separable extensions of k . Over a separably closed field it is k^X , and the P -action permutes its primitive idempotents. This is kX with its function-algebra structure. Compare [Bal15, Remark 4.6] and [BC18, Proposition 4.4]. $\square$

Lemma 2.7 (Transitive permutation modules). *For a finite p -group P , the permutation module $k(P/H)$ is indecomposable and is projective exactly when $H = 1$.*

Proof. Every nonzero P -module has a nonzero fixed vector, by nilpotence of the augmentation ideal. A transitive permutation module has a one-dimensional fixed space, so it cannot be a direct sum of two nonzero modules. For $H \neq 1$, its dimension is positive and less than $|P|$, hence it is not free. For $H = 1$ it is the regular module. $\square$

3. A COUNTEREXAMPLE FOR $\mathfrak{A}_7$

Throughout this section, k has characteristic 3. Let Ω , L , and b be as in Theorem 1.2, and put

$$\alpha = (123), \quad \beta = (456), \quad S = \langle \alpha, \beta \rangle \leq \mathfrak{A}_7.$$

Then $S \cong C_3 \times C_3$, and $|\mathfrak{A}_7| = 2520 = 9 \cdot 280$ shows that S is a Sylow 3-subgroup. In particular, $[\mathfrak{A}_7 : S]$ is invertible in k .

The proof has two parts. We first construct an orthogonal decomposition of $k\Omega$ and use its restriction to S to prove that the specified form $b : L \otimes L \rightarrow k$ has projective kernel and a section. These maps make the associativity and separability identities for $k \oplus L$ hold modulo projectives. We then prove that a permutation module whose restriction to S is stably a direct sum of copies of k has the same property over $\mathfrak{A}_7$. The equality $L^{\mathfrak{A}_7} = 0$ distinguishes our algebra from these permutation modules.

3.1. The orthogonal summand. Equip $k\Omega$ and $k\{1, \dots, 7\}$ with the symmetric bilinear forms for which their permutation bases, denoted by e_T and f_i , are orthonormal. Set

$$u = \sum_{T \in \Omega} e_T, \quad e = \sum_{i=1}^7 f_i, \quad W = e^\perp \subseteq k\{1, \dots, 7\}.$$

Lemma 3.1. *The submodule L is nondegenerate for the form on $k\Omega$, and*

$$k\Omega \cong k\{1, \dots, 7\} \oplus L, \quad \dim_k L = 28, \quad L^{\mathfrak{A}_7} = 0. \quad (3.1)$$

Proof. Since $\langle e, e \rangle = 7 = 1$ in k , there is an orthogonal decomposition $k\{1, \dots, 7\} = ke \oplus W$, with W nondegenerate. Consider the $\mathfrak{A}_7$ -linear incidence map

$$F : k\{1, \dots, 7\} \longrightarrow k\Omega, \quad F(f_i) = \sum_{T \ni i} e_T.$$

Let F^* be its adjoint. There are 15 triples containing a given point and 5 containing two given distinct points. Thus, with J the 7×7 all-ones matrix,

$$F^*F = 10I + 5J = I + 2J. \quad (3.2)$$

The map J vanishes on W , so $F|_W$ is an isometric embedding. Moreover,

$$F(e) = 3u = 0, \quad F^*(u) = 15e = 0, \quad \langle u, u \rangle = 35 = 2 \neq 0. \quad (3.3)$$

It follows that $D = ku \oplus F(W)$ is a nondegenerate submodule of dimension 7, and

$$D \cong k \oplus W \cong k\{1, \dots, 7\}.$$

Consequently $k\Omega = D \oplus D^\perp$.

Since $F(e) = 0$, the image of F is $F(W)$. Hence

$$D^\perp = u^\perp \cap \ker(F^*).$$

In coordinates these are precisely the equations defining L in (1.4). This proves the decomposition and the nondegeneracy of b , and gives $\dim_k L = 35 - 7 = 28$.

Finally, $\mathfrak{A}_7$ is transitive on Ω : a permutation carrying one triple to another can be made even, if necessary, by composing with a transposition in the complement of the target triple. Thus $(k\Omega)^{\mathfrak{A}_7} = ku$. Since $L \subseteq u^\perp$ and $\langle u, u \rangle \neq 0$, we obtain $L^{\mathfrak{A}_7} = 0$. $\square$

Lemma 3.2. *There is an ordinary kS -module isomorphism*

$$L \downarrow_S \cong k \oplus (kS)^{\oplus 3}. \quad (3.4)$$

Proof. The orbits of S on Ω are indexed by

$$(|T \cap \{1, 2, 3\}|, |T \cap \{4, 5, 6\}|, |T \cap \{7\}|).$$

A 3-cycle acts transitively on both the one-element and the two-element subsets of its support, and the actions on the two blocks are independent. Each row below is therefore one orbit.

type	orbit size	stabilizer in S
(3, 0, 0)	1	S
(0, 3, 0)	1	S
(2, 0, 1)	3	$\langle \beta \rangle$
(0, 2, 1)	3	$\langle \alpha \rangle$
(2, 1, 0)	9	1
(1, 2, 0)	9	1
(1, 1, 1)	9	1

Consequently,

$$(k\Omega) \downarrow_S \cong k^{\oplus 2} \oplus k[S/\langle \alpha \rangle] \oplus k[S/\langle \beta \rangle] \oplus (kS)^{\oplus 3},$$

$$(k\{1, \dots, 7\}) \downarrow_S \cong k \oplus k[S/\langle \alpha \rangle] \oplus k[S/\langle \beta \rangle].$$

Restricting (3.1) and applying ordinary direct-sum cancellation, Lemma 2.2, proves the claim. $\square$

Lemma 3.3. *The kernel Q of $b : L \otimes L \rightarrow k$ is projective. There is an $\mathfrak{A}_7$ -linear map $\gamma : k \rightarrow L \otimes L$ such that*

$$b\gamma = \text{id}_k, \quad L \otimes L = \gamma(k) \oplus Q. \quad (3.5)$$

In particular, this specific form induces an isomorphism $b : L \otimes L \simeq_{\text{st}} k$.

Proof. Choose a basis $v_1, \dots, v_{28}$ of L and its b -dual basis $v^1, \dots, v^{28}$, and define

$$\gamma(1) = c := \sum_{i=1}^{28} v_i \otimes v^i.$$

Under the identification $L \otimes L \cong \text{End}_k(L)$ induced by b , the tensor c corresponds to the identity. It is therefore basis-independent and $\mathfrak{A}_7$ -invariant. Symmetry of b makes c symmetric and gives the contraction identities

$$(\text{id}_L \otimes b)(\gamma \otimes \text{id}_L) = \text{id}_L, \quad (b \otimes \text{id}_L)(\text{id}_L \otimes \gamma) = \text{id}_L. \quad (3.6)$$

Furthermore,

$$b(c) = \sum_i b(v_i, v^i) = 28 = 1 \quad \text{in } k.$$

Thus $b\gamma = \text{id}_k$, giving the ordinary decomposition in (3.5).

It remains to prove that its complementary summand is projective. Lemma 3.2 and the tensor calculation in Lemma 2.1 give

$$\begin{aligned} (L \otimes L) \downarrow_S &\cong (k \oplus (kS)^{\oplus 3}) \otimes (k \oplus (kS)^{\oplus 3}) \\ &\cong k \oplus (kS)^{\oplus 6} \oplus (kS \otimes kS)^{\oplus 9} \\ &\cong k \oplus (kS)^{\oplus 87}. \end{aligned}$$

Here $kS \otimes kS$, with diagonal action, is free of rank 9. Restricting (3.5) and applying ordinary cancellation gives $Q \downarrow_S \cong (kS)^{\oplus 87}$. Since $[\mathfrak{A}_7 : S]$ is invertible in k , Lemma 2.1 implies that Q is projective. The difference $\text{id}_{L \otimes L} - \gamma b$ is its projection, so b and γ are inverse in the stable category. $\square$

3.2. A separable quadratic algebra. The multiplication (1.6) need not be associative as an ordinary bilinear map. The next lemma identifies the projective modules through which its associator and the two separability defects factor.

Lemma 3.4 (Quadratic construction). *Let k be a field of characteristic different from 2, and let K be a finite group. Let V be a finite-dimensional kK -module with a nondegenerate invariant symmetric bilinear form $b : V \otimes V \rightarrow k$. Suppose that*

$$(\dim_k V)1_k = 1_k, \quad Q := \ker(b) \text{ is projective.}$$

Then $B = k \oplus V$, with unit $1 = (1, 0)$ and multiplication

$$(a, x)(a', y) = (aa' + b(x, y), ay + a'x), \quad (3.7)$$

is a separable commutative algebra in $\text{stmod}(kK)$. If $\gamma : k \rightarrow V \otimes V$ is the inverse-form tensor and $c = \gamma(1)$, a bimodule section is represented by

$$\sigma(1) = \frac{1}{2}(1 \otimes 1 + c), \quad \sigma(x) = \frac{1}{2}(x \otimes 1 + 1 \otimes x) \quad (x \in V). \quad (3.8)$$

Proof. The inverse-form tensor satisfies the two contraction identities of (3.6), with V in place of L . The dimension hypothesis gives $b\gamma = \text{id}_k$. Thus

$$V \otimes V = \gamma(k) \oplus Q, \quad q := \text{id}_{V \otimes V} - \gamma b$$

is the projection onto the projective summand Q . The multiplication and unit are K -linear; the unit identities and commutativity hold as ordinary module maps.

The associator. Every component of the associator with a k -factor is zero. Its restriction to $V^{\otimes 3}$ is

$$\delta = b \otimes \text{id}_V - \text{id}_V \otimes b : V^{\otimes 3} \longrightarrow V.$$

The first contraction identity gives

$$\delta(\gamma \otimes \text{id}_V) = (b\gamma) \otimes \text{id}_V - (\text{id}_V \otimes b)(\gamma \otimes \text{id}_V) = 0.$$

It follows that $\delta = \delta(q \otimes \text{id}_V)$, so the associator factors through $Q \otimes V$, which is projective by Lemma 2.1. Hence the multiplication is associative in the stable category.

The multiplication section. The map σ in (3.8) is K -linear because c is invariant, and

$$\mu\sigma(1) = \frac{1}{2}(1 + b(c)) = 1, \quad \mu\sigma(x) = \frac{1}{2}(x + x) = x.$$

Thus $\mu\sigma = \text{id}_B$ already as an ordinary map. To prove separability, we must show that

$$(\mu \otimes \text{id}_B)(\text{id}_B \otimes \sigma) = \sigma\mu = (\text{id}_B \otimes \mu)(\sigma \otimes \text{id}_B) \quad (3.9)$$

in the stable category.

On $k \otimes k$, all three maps agree by the unit identities. On $k \otimes V$ and $V \otimes k$, the only additional equalities are

$$\begin{aligned} (\text{id}_B \otimes \mu)(\sigma(1) \otimes x) &= \frac{1}{2} \left(1 \otimes x + \sum_i v_i \otimes b(v^i, x)1 \right) = \sigma(x), \\ (\mu \otimes \text{id}_B)(x \otimes \sigma(1)) &= \frac{1}{2} \left(x \otimes 1 + \sum_i b(x, v_i)1 \otimes v^i \right) = \sigma(x), \end{aligned}$$

which follow from the contraction identities. On $V \otimes V$, for $x, y \in V$ we obtain

$$\begin{aligned} \sigma\mu(x \otimes y) &= \frac{1}{2}b(x, y)(1 \otimes 1 + c), \\ (\mu \otimes \text{id}_B)(\text{id}_B \otimes \sigma)(x \otimes y) &= \frac{1}{2}(b(x, y)1 \otimes 1 + x \otimes y), \\ (\text{id}_B \otimes \mu)(\sigma \otimes \text{id}_B)(x \otimes y) &= \frac{1}{2}(x \otimes y + b(x, y)1 \otimes 1). \end{aligned}$$

Subtracting either outer map in (3.9) from $\sigma\mu$ therefore gives zero on the first three summands and

$$\frac{1}{2}(\gamma b - \text{id}_{V \otimes V}) = -\frac{1}{2}q$$

on $V \otimes V$, followed by its inclusion in $B \otimes B$. Both differences factor through Q , proving the bimodule identities in the stable category. $\square$

Corollary 3.5. *The unit and multiplication in Theorem 1.2 make $A = k \oplus L$ a separable commutative algebra in $\text{stmod}(k\mathfrak{A}_7)$. Moreover,*

$$A \downarrow_S \cong k^{\oplus 2} \oplus (kS)^{\oplus 3}, \quad A \downarrow_{S \simeq_{\text{st}}} k^{\oplus 2}. \quad (3.10)$$

Proof. Apply Lemma 3.4 to $V = L$, using Lemmas 3.1 and 3.3 and $28 = 1$ in k . The restriction formula follows from Lemma 3.2. $\square$

3.3. Fixed points of permutation modules. We now prove the assertion about permutation modules needed to distinguish A from every kX .

Proposition 3.6. *Let $r \geq 0$ be an integer. If a finite $\mathfrak{A}_7$ -set X satisfies*

$$(kX) \downarrow_{S \simeq_{\text{st}}} k^{\oplus r},$$

then X has exactly r global fixed points, and the stabilizer of every point outside $X^{\mathfrak{A}_7}$ has order prime to 3. Consequently,

$$kX \cong k^{\oplus r} \oplus P \quad \text{for a projective } k\mathfrak{A}_7\text{-module } P. \quad (3.11)$$

The local input is that stable triviality determines the nonfree orbits of an S -set. The group-specific step will then propagate these fixed points among all subgroups generated by 3-cycles.

Lemma 3.7. *Let $r \geq 0$ be an integer, and let Z be a finite S -set such that $kZ \simeq_{\text{st}} k^{\oplus r}$. Then Z is the disjoint union of exactly r fixed points and free S -orbits. In particular,*

$$|Z^S| = r, \quad Z^P = Z^S \quad (1 \neq P \leq S). \quad (3.12)$$

Proof. By Lemma 2.7, each transitive permutation kS -module is indecomposable. Its dimension is 1, 3, or 9. Projective kS -modules are free by Lemma 2.4, so the first two types are nonprojective, while a module from a free orbit is the regular module kS . Decomposing Z into orbits and applying Lemma 2.2 excludes every orbit of size 3 and forces exactly r orbits of size 1. A nontrivial subgroup has no fixed points on a free S -orbit, proving (3.12). $\square$

Proof of Proposition 3.6. For each $T \in \Omega$, let C_T be the order-3 subgroup generated by a 3-cycle supported on T ; it does not depend on the orientation of the cycle. For disjoint $T, U \in \Omega$, set

$$S_{T,U} = \langle C_T, C_U \rangle \cong C_3 \times C_3.$$

This subgroup is conjugate to S inside $\mathfrak{A}_7$. Indeed, choose a permutation sending the two supports of S to T and U . If it is odd, compose on the right with a transposition in $\{1, 2, 3\}$, which normalizes S . The resulting conjugating permutation is even. Hence the hypothesis and Lemma 3.7 apply to every $S_{T,U}$, giving

$$X^{C_T} = X^{S_{T,U}} = X^{C_U}, \quad |X^{S_{T,U}}| = r \quad (T \cap U = \emptyset). \quad (3.13)$$

The graph on Ω in which disjoint triples are adjacent is connected. If T, T' differ by one element, their union has size 4, so its complement is a triple disjoint from both and gives a path of length 2. Any two triples can be joined by successive single-element replacements. Applying (3.13) along these paths yields a subset $Y \subseteq X$ of cardinality r with

$$X^{C_T} = Y \quad \text{for all } T \in \Omega. \quad (3.14)$$

Every 3-cycle therefore fixes Y pointwise. These cycles generate $\mathfrak{A}_7$: write an even permutation as a product of pairs of transpositions. A pair sharing one letter is a 3-cycle, an identical pair cancels, and for four distinct letters

$$(ab)(cd) = (acb)(acd).$$

Thus $Y \subseteq X^{\mathfrak{A}_7}$; the reverse inclusion follows from (3.14). We have proved $Y = X^{\mathfrak{A}_7}$.

An element of order 3 in $\mathfrak{A}_7$ is either a single 3-cycle or a product of two disjoint 3-cycles. In the first case its fixed set is Y . In the second case it generates a nontrivial subgroup of some $S_{T,U}$, whose fixed set is again Y by Lemma 3.7. Hence the stabilizer of any $x \in X \setminus Y$ has no element of order 3, and its order is prime to 3 by Cauchy's theorem. For the needed case of that theorem, cyclic rotation acts on the $|H|^2$ triples $(h_1, h_2, h_3) \in H^3$ satisfying $h_1 h_2 h_3 = 1$. Nonfixed orbits have size 3, and fixed triples correspond to elements h with $h^3 = 1$. If $3 \nmid |H|$, their number is divisible by 3; the identity solution forces a nonidentity solution of order 3.

Finally, for a subgroup $H \leq \mathfrak{A}_7$ of order prime to 3, the element

$$e_H = |H|^{-1} \sum_{h \in H} h \in kH$$

is an idempotent, and kHe_H is the trivial kH -module. It is a direct summand of kH , hence projective. Induction and Lemma 2.1 show that $k[\mathfrak{A}_7/H] \cong k\mathfrak{A}_7 \otimes_{kH} k$ is projective. Each orbit in $X \setminus Y$ has such a stabilizer. The decomposition

$$kX = kY \oplus k(X \setminus Y)$$

therefore proves (3.11). $\square$

3.4. Completion of the counterexample.

Proof of Theorem 1.2. Lemma 3.1 gives the dimension of L and the nondegeneracy of b . Corollary 3.5 supplies the separable commutative algebra structure on $A = k \oplus L$.

Suppose that $A \simeq_{\text{st}} kX$ for a finite $\mathfrak{A}_7$ -set X . Restriction preserves projectives by Lemma 2.1, so (3.10) gives $(kX) \downarrow_S \simeq_{\text{st}} k^{\oplus 2}$. Proposition 3.6 then implies

$$A \simeq_{\text{st}} kX \simeq_{\text{st}} k^{\oplus 2}. \quad (3.15)$$

We distinguish these stable objects by maps from the tensor unit. The trivial $k\mathfrak{A}_7$ -module is not projective: its restriction to S has dimension 1, whereas projective kS -modules are free by Lemma 2.4 and have dimension divisible by 9. Thus no nonzero scalar endomorphism of k factors through a projective module; otherwise its identity would factor through that module, making k projective. Therefore

$$\underline{\text{Hom}}_{k\mathfrak{A}_7}(k, k) = k.$$

On the other hand, Lemma 3.1 gives $\text{Hom}_{k\mathfrak{A}_7}(k, L) = L^{\mathfrak{A}_7} = 0$. Consequently

$$\underline{\text{Hom}}_{k\mathfrak{A}_7}(k, A) \cong k, \quad \underline{\text{Hom}}_{k\mathfrak{A}_7}(k, k^{\oplus 2}) \cong k^{\oplus 2},$$

contradicting (3.15). This argument holds over every field of characteristic 3, including $\overline{\mathbb{F}}_3$. $\square$

Remark 3.8. The multiplication (1.6) is not associative on the ordinary 29-dimensional vector space. Choose $0 \neq y \in L$, then $x \in L$ with $b(x, y) = 1$, and a nonzero $z \in \ker b(y, -)$; such a z exists since $\dim L > 1$. The ordinary associator on x, y, z is

$$b(x, y)z - b(y, z)x = z \neq 0.$$

Lemma 3.4 shows that this associator factors through $Q \otimes L$, and that both bimodule-linearity defects factor through Q . These projective factorizations are the reason the algebra exists in the stable category.

Remark 3.9. The counterexample also exists in the usual symmetric monoidal stable ∞ -categorical enhancement of $\text{stmod}(k\mathfrak{A}_7)$. Indeed, a separable commutative algebra in the homotopy category of a symmetric monoidal stable ∞ -category has an essentially unique separable commutative lift by [NPR26, Theorem 2.1], a form of [Ram23, Theorem B]. The algebra constructed above satisfies these hypotheses. An equivalence from its lift to a permutation algebra would descend to an isomorphism in the homotopy category, contradicting Theorem 1.2.

Remark 3.10. The form identifies L with its dual, and Lemma 3.3 gives $L \otimes L^* \cong k \oplus Q$ with Q projective. Thus L is endotrivial, and (3.4) says that it is Sylow-trivial. Related modules for symmetric and alternating groups are discussed in [Gro23, §6.3, especially the case $n = 7, p = 3$]. The proof here uses neither their classification nor a computation of a categorical Galois group. Since $\mathfrak{A}_7$ is not a 3-group, this counterexample is consistent with the classification for finite p -groups.

4. GRAPH IDEMPOTENTS AND SPLITTING TOWERS

An algebra A in an additive symmetric monoidal category with unit $\mathbf{1}$ is *connected* if the ring $\text{Hom}(\mathbf{1}, A)$ has no idempotents other than 0 and 1. In the categorical calculations below, the category is additive and idempotent-complete, and its tensor product is additive in each variable. Formulae with variables stand for identities of morphisms from tensor products; no detection of morphisms by maps from the unit is assumed.

Lemma 4.1. *For a finite p -group P , every separable commutative algebra in $\text{stmod}(kP)$ is a finite product of nonzero connected separable algebras. The empty product is the zero algebra.*

Proof. The ring $\text{Hom}_{\text{stmod}(kP)}(k, A)$ is finite-dimensional and commutative, so its unit is a finite sum of orthogonal primitive idempotents. An idempotent $a : k \rightarrow A$ induces an idempotent A -linear endomorphism of A by multiplication. Splitting these endomorphisms gives mutually orthogonal ideals, with units the corresponding idempotents, and hence a product decomposition of A . The bimodule section of multiplication restricts to each factor, so every factor is separable. Its ring of maps from k is the factor of $\text{Hom}_{\text{stmod}(kP)}(k, A)$ cut out by the corresponding primitive idempotent; it has no further nontrivial idempotents. A nonzero idempotent cannot give the zero algebra, since it is the unit of its factor. $\square$

Lemma 4.2 (Graph idempotents). *Let $A \neq 0$ be a connected separable commutative algebra in an idempotent-complete symmetric monoidal additive category $\mathcal{C}$ whose tensor product is additive in each variable. Choose a bimodule section σ of multiplication and set $e = \sigma\eta$. For each algebra automorphism α , set $e_\alpha = (1 \otimes \alpha)e$. Then*

$$\alpha \neq \beta \implies e_\alpha e_\beta = 0. \quad (4.1)$$

If $F : \mathcal{C} \rightarrow \mathcal{D}$ is an additive strong symmetric monoidal functor to another such category and $F(A) \neq 0$, then $F(\alpha) = 1$ implies $\alpha = 1$.

Proof. Bimodule linearity and the section identity give

$$e^2 = e, \quad \mu e = \eta, \quad (a \otimes 1)e = (1 \otimes a)e, \quad \sigma(a) = (a \otimes 1)e. \quad (4.2)$$

For the first identity, move one factor of e across the other using the third identity and contract the resulting pair with $\mu e = \eta$. Multiplication identifies the ideal $(A \otimes A)e$, defined by splitting multiplication by e , with A . Its inverse sends a to $(a \otimes 1)e$. Twisting the second tensor factor therefore gives an algebra isomorphism

$$(A \otimes A)e_\alpha \xrightarrow{\mu(1 \otimes \alpha^{-1})} A, \quad a \longmapsto (a \otimes 1)e_\alpha \quad \text{for its inverse.}$$

The unit of the ideal on the left is e_α .

The product $e_\alpha e_\beta$ is an idempotent in this copy of A . Connectedness makes it either zero or e_α . In the latter case, multiply

$$(a \otimes 1)e_\beta = (1 \otimes \beta(a))e_\beta$$

by e_α and apply $\mu(1 \otimes \alpha^{-1})$. The result is $a = \alpha^{-1}\beta(a)$ as an identity of morphisms $A \rightarrow A$. Thus $\alpha = \beta$, proving (4.1).

Suppose that $F(\alpha) = 1$ but $\alpha \neq 1$. Then $F(e_\alpha) = F(e_1)$. Additivity of F makes the image of (4.1) a zero product. Since $F(e_1)$ is idempotent, it follows that $F(e_1) = 0$. But $F(\mu)F(e_1) = F(\eta)$ is the unit of $F(A)$, so its unit is zero. The unit identity would then imply $F(A) = 0$, a contradiction. $\square$

For an abelian p -group P , the action of every $g \in P$ on a commutative algebra A in $\text{stmod}(kP)$ is a P -linear algebra automorphism. Define its *stable action kernel* by

$$K_A = \ker(P \longrightarrow \text{Aut}_{\text{CAlg}(\text{stmod}(kP))}(A)). \quad (4.3)$$

This definition requires triviality in the stable category; it does not assert actual triviality on a core representative.

Lemma 4.3. *If $P \neq 1$ is abelian and $A \neq 0$ is connected and separable in $\text{stmod}(kP)$, then $K_A \neq 1$.*

Proof. If $K_A = 1$, the graph idempotents e_g of the actions of the elements of P are pairwise orthogonal by Lemma 4.2. Choose a core representative of A . Lemma 2.5 gives

$$\sum_{g \in P} e_g = (1 \otimes N_P)e = 0.$$

Multiplying by e_1 yields $e_1 = 0$, whereas $\mu e_1 = \eta \neq 0$. The last inequality follows from the unit identity and $A \neq 0$. $\square$

For a separable commutative algebra A in $\text{stmod}(kG)$, its splitting tower begins with $A^{[0]} = k$ and $A^{[1]} = A$ and is characterized by

$$A^{[j]} \otimes_{A^{[j-1]}} A^{[j]} \simeq A^{[j]} \times A^{[j+1]},$$

with multiplication as projection onto the first factor. The splitting theorem and its uniqueness give these terms successively [Bal14, Theorem 2.4 and Definition 3.1]. Each term is again a separable commutative algebra in the original category [Bal14, Remark 1.5]. The *degree* of A is the least $d \geq 0$ such that $A^{[d+1]} = 0$, and is infinite if no such d exists. In particular, $\deg(0) = 0$, including when the ambient category is zero. Splitting towers commute with symmetric monoidal exact functors [Bal14, Theorem 3.7(a)].

Proposition 4.4 (A finite degree bound). *For any finite group G over a field of characteristic $p > 0$, every separable commutative algebra A in $\text{stmod}(kG)$ has finite degree, with*

$$\deg(A) \leq c(A). \quad (4.4)$$

Proof. If $A = 0$, the assertion follows from the convention above. Otherwise, the splitting-tower identity of [Bal14, Theorem 3.9(b)], after forgetting the module structure, gives

$$A \otimes A^{[j]} \simeq (A^{[j]})^{\oplus j} \oplus A^{[j+1]} \quad (j \geq 1). \quad (4.5)$$

Set $m = c(A) \geq 1$. The additivity and tensor inequality in Lemma 2.3, which hold for every finite group, imply

$$j c(A^{[j]}) + c(A^{[j+1]}) \leq m c(A^{[j]}).$$

Taking $j = m$ gives $c(A^{[m+1]}) \leq 0$. Since core dimensions are nonnegative, $A^{[m+1]}$ is zero in the stable category and $\deg(A) \leq m$. $\square$

5. A RELATIVE NORM ON INVARIANT VECTORS

Let $P = H \times Q$ be an abelian p -group and let M be a core representative of a nonzero connected separable algebra A . When $H \cap K_A = 1$, the construction below equips M^H with a separable algebra structure over Q . If M is free over H and $Q \neq 1$, this algebra is still nonzero and connected, and its core dimension is $\dim_k M/|H|$. These are the assertions needed to reduce the rank of an elementary abelian group. The restricted multiplication descends because stable errors factor through the invariants of projective modules. The relative norm supplies a bimodule section of that multiplication; its section identity follows from the orthogonality of the H -graph idempotents.

Proposition 5.1 (An invariant separable algebra). *Let $P = H \times Q$ be an abelian finite p -group, and let $A \neq 0$ be a connected separable commutative algebra in $\text{stmod}(kP)$ with $H \cap K_A = 1$. On a core representative M , choose module-map representatives μ, η, σ of multiplication, the unit, and a bimodule section. Then $C = M^H$, with the restricted multiplication and unit, is a separable commutative algebra in $\text{stmod}(kQ)$. A section is represented by*

$$\sigma_H = (N_H \otimes 1)\sigma|_{M^H} : M^H \longrightarrow M^H \otimes M^H. \quad (5.1)$$

For $Q = 1$, the conclusion is interpreted in the zero stable category.

Proof. The algebra on C . There is an ordinary Q -module isomorphism $(kP)^H \cong kQ$, sending N_{Hq} to q . Taking H -invariants therefore sends projective P -modules to projective Q -modules and induces an additive functor

$$(-)^H : \text{stmod}(kP) \longrightarrow \text{stmod}(kQ). \quad (5.2)$$

Only additivity is used here; exactness and a strong monoidal structure on this functor are not required.

Equivariance makes M^H closed under the chosen multiplication and puts the unit in M^H . Suppose that the error in an algebra identity factors as $U \rightarrow F \rightarrow V$ with F projective over P . Its restriction

to an H -trivial submodule of U factors through F^H , which is projective over Q . Applying this observation to inputs in C , $C \otimes C$, and $C^{\otimes 3}$ proves the unit, commutativity, and associativity identities in $\text{stmod}(kQ)$. Write μ_C for the restricted multiplication.

The norm section. Define

$$r : (M \otimes M)^H \longrightarrow M^H \otimes M^H, \quad r(z) = (N_H \otimes 1)z.$$

The value is invariant under H in the first factor. It is also invariant under the diagonal action, since $N_H \otimes 1$ commutes with that action. These two invariances imply invariance in the second factor. Over a field, the simultaneous invariant subspace is $M^H \otimes M^H$, proving that r has the stated target. The map is Q -linear because Q commutes with H . Since σ carries M^H into $(M \otimes M)^H$, the displayed σ_H is $r\sigma|_{M^H}$.

Let α_h denote the stable action of $h \in H$ on A , and set

$$T_h = \mu(\alpha_h \otimes 1)\sigma : A \longrightarrow A.$$

For $h = 1$, this is the identity. If $h \neq 1$, then $\alpha_h \neq 1$ by $H \cap K_A = 1$. Lemma 4.2 gives $e_1 e_{\alpha_h^{-1}} = 0$. Applying the algebra map μ and using $\mu e_1 = \eta$ gives $\mu e_{\alpha_h^{-1}} = 0$. Applying α_h now gives $\mu(\alpha_h \otimes 1)e = 0$. Thus

$$T_h(a) = \alpha_h(a) \mu(\alpha_h \otimes 1)e = 0,$$

where we used (4.2). Summing over H yields

$$\mu(N_H \otimes 1)\sigma = \sum_{h \in H} T_h = \text{id}_A \quad \text{in } \text{stmod}(kP). \quad (5.3)$$

Its restriction to M^H is $\mu_C \sigma_H = \text{id}_C$ in $\text{stmod}(kQ)$, since the error factors through F^H whenever the original error factors through a projective P -module F .

Bimodule linearity. For $a \in M^H$ and $x \in M$, ordinary equivariance of μ gives

$$N_H \mu(a, x) = \mu(a, N_H x), \quad N_H \mu(x, a) = \mu(N_H x, a). \quad (5.4)$$

It follows that r commutes with the outer left and right actions of C on $(M \otimes M)^H$. Restrict each of the two bimodule identities for σ to $C \otimes C$ and compose with r . The identities (5.4) turn their two right-hand sides into

$$(\mu_C \otimes 1)(1 \otimes \sigma_H) \quad \text{and} \quad (1 \otimes \mu_C)(\sigma_H \otimes 1),$$

respectively, while the left-hand side is $\sigma_H \mu_C$. If an original error factors through a projective P -module F , its restriction factors through $F^H \rightarrow (M \otimes M)^H$, and we can compose this latter map with r . The new error consequently still factors through the projective Q -module F^H . Both bimodule identities therefore hold in $\text{stmod}(kQ)$, proving separability. $\square$

Lemma 5.2. *In Proposition 5.1, suppose that $Q \neq 1$ and $M \downarrow_H$ is free. Then $C = M^H$ is a nonzero core Q -module, the resulting algebra is connected, and*

$$\dim_k C = \frac{\dim_k M}{|H|}. \quad (5.5)$$

Proof. For a free H -module the invariant subspace is its norm image, and each regular summand contributes one dimension. Hence $C = N_H M$ and (5.5) holds. Moreover,

$$N_Q C = N_Q N_H M = N_P M = 0$$

by Lemma 2.5. The norm acts nontrivially on any nonzero free Q -module, so C has no projective direct summand. Its dimension is positive, and it is therefore nonzero in the stable category as well.

Finally, there are identifications

$$C^Q = M^P, \quad \underline{\text{Hom}}_{kQ}(k, C) = C^Q, \quad \underline{\text{Hom}}_{kP}(k, M) = M^P.$$

The second and third use the core assertion and Lemma 2.5. These are ring identifications, since the multiplication on C is the restriction of that on M . The last ring has no nontrivial idempotents by connectedness of A , and the same is true of $\underline{\mathrm{Hom}}_{kQ}(k, C)$. $\square$

6. SPECTRAL INPUTS AND TORUS DESCENT

Let P be a finite p -group, and let $\mathrm{StMod}(kP)$ denote the presentable symmetric monoidal stable ∞ -category enhancing the stable category of all kP -modules. Its tensor unit is a compact generator, and there is a symmetric monoidal equivalence

$$\mathrm{StMod}(kP) \simeq \mathrm{Mod}(k^{tP}).$$

The dualizable objects on either side are the compact objects. Thus the finite stable module category is the homotopy category of $\mathrm{Perf}(k^{tP})$. These identifications follow from [MatT, Theorem 2.9] and [Mat16, Theorem 2.30]. In spectral formulas, a field denotes its Eilenberg–Mac Lane commutative ring spectrum.

Proposition 6.1 (Separable lifting and limits). *Let $\mathcal{C}$ be an idempotent-complete symmetric monoidal stable ∞ -category whose tensor product is exact in each variable. Passage to the homotopy category induces an equivalence between its category of separable commutative algebras and the category of separable commutative algebras in $\mathrm{Ho}(\mathcal{C})$. In particular, the former has discrete mapping spaces.*

The functor of separable commutative algebras preserves limits of symmetric monoidal stable ∞ -categories and exact symmetric monoidal functors. Dualizability is detected componentwise in these limits, so the same assertion holds after restricting to dualizable underlying objects.

Proof. The lifting and mapping-space assertions are [NPR26, Theorem 2.1], in the stated form of [Ram23, Theorem B]. Limit preservation is [NPR26, Proposition 2.3]. A symmetric monoidal functor preserves duals, evaluation, and coevaluation. Since the choices of this duality data are contractible when they exist, compatible componentwise duals give a dual in the limit; compare [NPR26, Remark 4.2]. $\square$

For $r \geq 1$, let $k^{t\mathbb{T}^r}$ be the torus Tate ring. By definition, it is the localization of $C^*(B\mathbb{T}^r; k)$ away from the iterated cofiber of its r degree-two cohomology generators [MatT, Definition 3.10]. We use the cohomological grading

$$S_r(k) = k[x_1, \dots, x_r] = \pi_{-*}C^*(B\mathbb{T}^r; k), \quad |x_i| = 2.$$

When taking Proj , we instead give each x_i polynomial degree one.

Proposition 6.2 (Torus action and geometric model). *Let $E = C_p^r$, and choose an exact sequence $0 \rightarrow L' \rightarrow L \rightarrow E \rightarrow 0$ with $L \cong \mathbb{Z}^r$. There is a symmetric monoidal action of $BL \simeq \mathbb{T}^r$ on $\mathrm{StMod}(kE)$ whose monodromy is the action induced by $L \rightarrow E$. It satisfies*

$$\mathrm{StMod}(kE)^{hBL} \simeq \mathrm{Mod}(k^{tBL'}) \simeq \mathrm{Mod}(k^{t\mathbb{T}^r}). \quad (6.1)$$

The extension $k^{t\mathbb{T}^r} \rightarrow k^{tE}$ is faithful; in particular, extension of scalars is conservative.

There is an even-periodic derived scheme $\mathfrak{X}_r$ with underlying ordinary scheme $X_r = \mathbb{P}_k^{r-1}$ and a symmetric monoidal equivalence

$$\mathrm{Mod}(k^{t\mathbb{T}^r}) \simeq \mathrm{QCoh}(\mathfrak{X}_r). \quad (6.2)$$

The standard affine charts $U_i = D_+(x_i)$ have ring spectra

$$\begin{aligned} R_i &= C^*(B\mathbb{T}^r; k)[x_i^{-1}] \simeq (k^{t\mathbb{T}^r})[x_i^{-1}], \\ \pi_0 R_i &= k[x_1/x_i, \dots, \widehat{x_i/x_i}, \dots, x_r/x_i]. \end{aligned} \quad (6.3)$$

These ring spectra are even periodic, with regular Noetherian π_0 of Krull dimension $r - 1$.

Proof. The action is constructed in [MatT, Construction 3.5]; its monodromy agrees with the group action on representations. The faithful torus-Galois extension and fixed-point equivalence are [MatT, Construction 3.11 and Theorem 3.12].

The geometric model and its affine charts are constructed in the proof of [MatT, Proposition 5.4]; see also [Mat16, §9.4]. Inverting x_i already kills the iterated quotient by all the x_j , so localizing the torus Tate ring at x_i agrees with localizing the cochain ring there. This gives the second description of R_i . Its periodicity and the displayed formula for $\pi_0 R_i$ follow from the polynomial cohomology calculation. These constructions apply in every positive characteristic. $\square$

The equivalence (6.2), sometimes called 0-affineness, concerns quasi-coherent modules on the derived scheme; the ordinary scheme X_r need not be affine.

Lemma 6.3. *Suppose that a separable commutative algebra $A \in \text{stmod}(kE)$ has $K_A = E$. Then A is the image under the forgetful functor in (6.1) of an algebra*

$$\tilde{A} \in \text{CAlg}^{\text{sep}}(\text{Perf}(k^{t\mathbb{T}^r})).$$

Proof. An S^1 -action on an ordinary category is a natural automorphism of its identity functor. The homotopy-fixed category is the full subcategory of objects on which this automorphism is the identity [MatT, Example 3.2]. Applying this description to each circle factor of a torus gives the corresponding statement for all monodromies.

By Proposition 6.1, the separable commutative algebras in $\text{StMod}(kE)$ form an ordinary category. The monodromies of A in this category are trivial: they are induced by $L \rightarrow E$, and $K_A = E$ is precisely triviality of the stable action by algebra automorphisms. Thus A has a homotopy-fixed structure as a separable commutative algebra. Limit preservation in Proposition 6.1 identifies it with a separable commutative algebra in $\text{StMod}(kE)^{hBL}$. Its underlying object is dualizable by the componentwise criterion in the same proposition. Under (6.1), it is therefore perfect. $\square$

Remark 6.4. Trivial monodromy in the homotopy category does not by itself give such a lift for an arbitrary stable object. Here the discreteness of the mapping spaces of separable commutative algebras removes the higher coherence obstruction.

Lemma 6.5. *Every separable commutative perfect $k^{t\mathbb{T}^r}$ -algebra has finite splitting-tower degree.*

Proof. Extend scalars along the faithful extension $k^{t\mathbb{T}^r} \rightarrow k^{tE}$ of Proposition 6.2. Perfect modules remain perfect, so their images correspond to finite stable E -modules. Proposition 4.4 makes the image of the given algebra have finite degree. Splitting towers commute with this exact symmetric monoidal functor [Bal14, Theorem 3.7(a)]. Conservativity therefore makes the same tower term vanish before extension of scalars. $\square$

For a perfect $k^{t\mathbb{T}^r}$ -module M , let M_i denote its restriction to R_i . The $\pi_0 R_i$ -module $\pi_0 M_i \oplus \pi_1 M_i$ is finitely generated: this is true for finite cell modules and is preserved by retracts, using Noetherianity. The resulting closed supports agree on overlaps and define

$$\text{Supp}(M) \subseteq X_r.$$

The chart description and (6.2) show that $\text{Supp}(M) = \emptyset$ if and only if $M = 0$.

Proposition 6.6 (Local spectral inputs). *Let R be an even-periodic commutative ring spectrum with $\pi_0 R$ regular Noetherian of finite Krull dimension.*

- (1) *If $\pi_0 R$ is local with regular parameters $t_1, \dots, t_d$, and M is perfect with $\pi_1(M/(t_1, \dots, t_d)) = 0$, then $\pi_1 M = 0$ and $\pi_0 M$ is a finite free $\pi_0 R$ -module.*
- (2) *A full-support perfect R -module generates $\text{Perf}(R)$ as a thick subcategory. In particular, a full-support perfect commutative R -algebra is descendable.*
- (3) *Every finite cover of R is the spectral lift of a finite étale $\pi_0 R$ -algebra. If 2 is invertible in $\pi_0 R$, every separable commutative perfect R -algebra is a finite cover.*

Proof. Part (1) is [Mat16, Lemma 6.32]. The quotient in its hypothesis is an iterated module cofiber; no commutative algebra structure on that quotient is required. Part (2) follows from [Mat15, Proposition 2.11 and Theorem 2.13]. The first assertion of (3) is [Mat16, Theorem 6.29], which includes characteristic two. The assertion for general separable commutative perfect algebras when 2 is invertible is [NP24, Theorem 10.7]. $\square$

Here a commutative algebra D is *descendable* if the thick tensor ideal it generates contains the unit. We use the following sufficient case of the finite-cover condition: if D is descendable and

$$D \otimes_R B \simeq D^{\times n} \quad \text{as } D\text{-algebras,}$$

then B is a *finite cover* of R [Mat16, Definition 6.1]. A spectral étale algebra has ordinary étale π_0 and satisfies the canonical flat-base-change formula for all homotopy groups. Such lifts, and all maps between them, are uniquely determined by π_0 [Mat16, Theorem 2.32].

Theorem 6.7. *Let k be separably closed of characteristic $p > 0$, and let $r \geq 1$. Every separable commutative perfect $k^{t\mathbb{T}^r}$ -algebra is a finite product of copies of $k^{t\mathbb{T}^r}$.*

In odd characteristic, the local assertion follows directly from Proposition 6.6(3). In characteristic two, that argument does not apply to an arbitrary separable algebra. We will first show that a nonzero algebra has a point of support defined over $\mathbb{F}_2$. At such a point a map of tori supplies a commutative residue algebra. Separability then forces its residue fiber to have only even homotopy, and local freeness propagates the support to the generic point. Applied to the last nonzero splitting-tower layer, this full-support assertion produces a descendable algebra that splits the original algebra. The resulting local finite covers are étale and glue on projective space. The proof of the theorem is completed in Section 9.

7. CHARACTERISTIC TWO: A RATIONAL POINT IN THE SUPPORT

Throughout this section, k is algebraically closed of characteristic 2. The support assertion below applies to all perfect commutative algebras, without a separability hypothesis.

Lemma 7.1 (The square operations needed here). *For an $\mathbb{E}_\infty$ - $H\mathbb{F}_2$ -algebra B , there are natural additive operations*

$$\widetilde{\text{Sq}}^j : \pi_{-n}B \longrightarrow \pi_{-n-j}B \quad (n, j \in \mathbb{Z}).$$

They vanish for $j > n$, have $\widetilde{\text{Sq}}^n(z) = z^2$ on $\pi_{-n}B$, and satisfy the Cartan formula with integer indices:

$$\widetilde{\text{Sq}}^m(zw) = \sum_{i \in \mathbb{Z}} \widetilde{\text{Sq}}^i(z) \widetilde{\text{Sq}}^{m-i}(w).$$

The sum is finite: for classes of cohomological degrees n and q , only $m - q \leq i \leq n$ can contribute.

On $S_r(k) = \pi_{-}C^*(B\mathbb{T}^r; k)$, define*

$$\varphi\left(\sum_{\alpha} a_{\alpha} x^{\alpha}\right) = \sum_{\alpha} a_{\alpha}^2 x^{\alpha},$$

and define k -linear operators D_j by

$$f(x_1 + tx_1^2, \dots, x_r + tx_r^2) = \sum_{j \geq 0} D_j(f) t^j. \quad (7.1)$$

Then, for homogeneous $f \in S_r(k)$ and $j \geq 0$,

$$\widetilde{\text{Sq}}^0(f) = \varphi(f), \quad \widetilde{\text{Sq}}^{2j}(f) = \varphi(D_j(f)). \quad (7.2)$$

Naturality of these formulas is valid for maps into nonconnective commutative ring spectra.

Proof. These are the cohomologically reindexed power operations; see [May70, Definition 2.2, Proposition 2.3, Corollary 2.7, and §5]. More explicitly, a representative $\Sigma^{-n}H\mathbb{F}_2 \rightarrow B$ induces a map from its second extended power in $\text{Mod}(H\mathbb{F}_2)$,

$$((\Sigma^{-n}H\mathbb{F}_2) \otimes_{H\mathbb{F}_2} (\Sigma^{-n}H\mathbb{F}_2))_{hC_2} \longrightarrow B.$$

Here C_2 interchanges the two tensor factors. The class in $H_{n-j}(BC_2; \mathbb{F}_2)$ gives $\widetilde{\text{Sq}}^j$ when $n - j \geq 0$, and the operation is zero when $n - j < 0$. This construction requires no connectivity assumption. The degree-zero class in the extended power gives the square, and the coproduct on $H_*(BC_2; \mathbb{F}_2)$ gives the displayed Cartan formula.

On the canonical $\mathbb{F}_2$ -cochains of a space these operations are the Steenrod squares. Negative squares vanish, and $\widetilde{\text{Sq}}^0$ is the identity [May70, Theorem 7.9 and Proposition 8.1]. Thus the only nonzero operations on a degree-two generator of $H^*(B\mathbb{T}^r; \mathbb{F}_2)$ are

$$\widetilde{\text{Sq}}^0(x_i) = x_i, \quad \widetilde{\text{Sq}}^2(x_i) = x_i^2.$$

On a scalar $a \in k$, the only operation that can be nonzero is $\widetilde{\text{Sq}}^0(a) = a^2$, by the square formula and the concentration of π_*Hk in degree zero. The coefficient-extension map, additivity, and Cartan therefore give (7.2). In particular, the use of integer indices on a general target leaves these source formulas unchanged. We do not assert that $\widetilde{\text{Sq}}^0$ is the identity on an arbitrary algebra or on its coefficient field. $\square$

Lemma 7.2 (A rational support point). *Let $B \neq 0$ be an $\mathbb{E}_\infty$ - $k^{t\mathbb{T}^r}$ -algebra whose underlying module is perfect. Then*

$$\text{Supp}(B) \cap \mathbb{P}^{r-1}(\mathbb{F}_2) \neq \emptyset. \quad (7.3)$$

Proof. The unit of B defines a homogeneous ideal

$$I = \ker(S_r(k) \longrightarrow \pi_{-*}B).$$

A homogeneous scalar annihilates the homotopy of a unital algebra if and only if it annihilates the unit: its action on any class is multiplication by its value on the unit. By (6.3), $B_i \simeq B[x_i^{-1}]$, so homotopy localizes. Its graded annihilator over $S_r(k)[x_i^{-1}]$ is $I[x_i^{-1}]$. By periodicity, the annihilator over the ordinary ring $\pi_0 R_i$ is

$$\text{Ann}_{\pi_0 R_i}(\pi_0 B_i \oplus \pi_1 B_i) = (I[x_i^{-1}])_0.$$

This degree-zero ideal cuts out $V_+(I) \cap U_i$. Finite generation of the homotopy modules on the charts therefore gives

$$\text{Supp}(B) = V_+(I). \quad (7.4)$$

This projective zero locus is nonempty because $B \neq 0$ and support detects zero objects. No finite-generation assertion for π_*B over $S_r(k)$ before localization is needed.

Naturality of $\widetilde{\text{Sq}}^0$ gives $\varphi(I) \subseteq I$. For each polynomial degree d , the space I_d is finite-dimensional, and φ is a Frobenius-semilinear bijection on $S_r(k)_d$. Since k is perfect, $\varphi(I_d)$ is a k -subspace of the same dimension as I_d . The inclusion is therefore an equality. Thus $\varphi(I) = I$, and the other operations in (7.2) give

$$D_j(I) \subseteq I \quad (j \geq 0). \quad (7.5)$$

Let $V = V(I) \subseteq k^r$. Substitution in (7.1), together with (7.5), shows that

$$a \in V, t \in k \implies a + t(a_1^2, \dots, a_r^2) \in V. \quad (7.6)$$

Since k is algebraically closed and $V_+(I)$ is nonempty, V contains a nonzero vector. Choose one with the least possible number of nonzero coordinates. If two nonzero coordinates a_i, a_j were distinct, taking $t = a_i^{-1}$ in (7.6) would kill the i -th coordinate while leaving the j -th coordinate $a_j(1 + a_j/a_i)$ nonzero. All zero coordinates would remain zero, contradicting minimality. Hence all

nonzero coordinates are equal. By homogeneity of I , scaling gives a nonzero vector in $\{0, 1\}^r$, as required. $\square$

8. CHARACTERISTIC TWO: RESIDUE ALGEBRAS AND FULL SUPPORT

We continue with k algebraically closed of characteristic two. The point furnished by Lemma 7.2 admits a commutative residue algebra induced by a map of tori. This algebra structure is needed to apply separability to the residue fiber.

Lemma 8.1 (Residue algebra at a rational point). *Let $y \in \mathbb{P}^{r-1}(\mathbb{F}_2)$, viewed as a closed point of $\mathbb{P}_k^{r-1}$, and let R_y be the local ring spectrum of $\mathfrak{X}_r$ at y . There is an $\mathbb{E}_\infty$ - R_y -algebra K_y such that*

$$\pi_* K_y \cong k[u^{\pm 1}], \quad |u| = -2. \quad (8.1)$$

For suitable regular parameters $t_2, \dots, t_r$ of $\pi_0 R_y$, its underlying module satisfies

$$K_y \simeq R_y / (t_2, \dots, t_r). \quad (8.2)$$

The right-hand side denotes iterated cofibers of multiplication.

Proof. After permuting coordinates, write $y = [1 : a_2 : \dots : a_r]$ with $a_j \in \mathbb{F}_2$. The coordinate change $x'_1 = x_1$, $x'_j = x_j - a_j x_1$ is the reduction of an integral unimodular change of coordinates and hence is induced by an automorphism of the torus. It suffices to treat $y = [1 : 0 : \dots : 0]$.

Inclusion of the first torus factor induces an $\mathbb{E}_\infty$ -map

$$C^*(B\mathbb{T}^r; k)[x_1^{-1}] \longrightarrow C^*(B\mathbb{T}; k)[u^{-1}], \quad x_1 \longmapsto u, \quad x_j \longmapsto 0 \ (j > 1).$$

Every degree-zero function nonvanishing at y maps to a nonzero scalar. The map therefore extends to R_y . Define K_y to be its target, whose homotopy is (8.1).

For the module identification, the external cochain product gives an equivalence over Hk ,

$$\bigotimes_{j=1}^r C^*(B\mathbb{T}; k) \xrightarrow{\sim} C^*(B\mathbb{T}^r; k).$$

Indeed, on homotopy it is the Künneth isomorphism; cohomology is of finite type and bounded below, so each degree involves a finite sum. For one factor, the augmentation $C^*(B\mathbb{T}; k) \rightarrow k$ identifies k with the cofiber of multiplication by its degree-two cohomology generator. The generator maps to zero under augmentation, providing the comparison map from the cofiber to k , and its long exact homotopy sequence proves that this map is an equivalence.

Tensor these cofiber descriptions for factors 2 through r , invert x_1 , and localize at y . Since x_1 is a periodicity unit, the quotients by x_j identify with the quotients by the degree-zero parameters $t_j = x_j/x_1$. This proves (8.2). When $r = 1$, the parameter list is empty and $K_y = R_y$. $\square$

Lemma 8.2 (Odd classes vanish after a graded-field base change). *Let k have characteristic two. A finite-dimensional $\mathbb{Z}/2$ -graded separable commutative k -algebra has zero odd part. Consequently, if K is an $\mathbb{E}_\infty$ -ring with $\pi_* K = k[u^{\pm 1}]$, $|u| = -2$, then every separable commutative perfect K -algebra D has $\pi_{\text{odd}} D = 0$.*

Proof. In characteristic two, forgetting the grading is symmetric monoidal. The underlying ordinary algebra of a graded separable commutative algebra $T = T_0 \oplus T_1$ is therefore finite étale over k . Define $\partial : T \rightarrow T$ to be zero on T_0 and the identity on T_1 . It is a derivation: in the odd-odd case the right-hand side of the Leibniz rule is twice the product and hence zero. A finite étale algebra has no nonzero k -derivations. For instance, after a faithfully flat extension to a separable closure it is a product of copies of the base field, where a derivation vanishes on scalars and idempotents. Thus $\partial = 0$, so $T_1 = 0$.

When $\pi_* K$ is a graded field, the Künneth map is an isomorphism for perfect K -modules. Homotopy groups therefore give a strong symmetric monoidal functor to graded $\pi_* K$ -modules. Setting $u = 1$

and reducing degrees modulo two is again symmetric monoidal. It sends D to a finite-dimensional graded separable commutative k -algebra, so the first assertion gives the result. $\square$

Proposition 8.3 (Full support). *Over an algebraically closed field k of characteristic two, every nonzero separable commutative perfect $k^{t^{\mathbb{T}^r}}$ -algebra B satisfies*

$$\mathrm{Supp}(B) = \mathbb{P}_k^{r-1}. \quad (8.3)$$

Proof. Choose a rational support point y by Lemma 7.2; thus $B_y \neq 0$. Lemma 8.1 provides a commutative residue algebra K_y whose underlying R_y -module is the quotient by a regular parameter sequence.

The residue fiber $B_y \otimes_{R_y} K_y$ is nonzero. To see this, suppose the iterated quotient were zero. The long exact homotopy sequence for the last parameter would make multiplication by that parameter surjective on the even and odd homotopy modules of the preceding quotient. Those modules are finitely generated over the local ring $\pi_0 R_y$, so Nakayama's lemma makes them zero. Repeating the argument for the preceding parameters would give $B_y = 0$.

As a commutative algebra, this residue fiber is still separable. Lemma 8.2 therefore makes its odd homotopy vanish. Proposition 6.6(1) now gives

$$\pi_1 B_y = 0, \quad \pi_0 B_y \text{ is a nonzero finite free } \pi_0 R_y\text{-module.}$$

It remains nonzero after localization to the generic point of $\mathbb{P}_k^{r-1}$. The closed support of B thus contains that generic point, and irreducibility gives (8.3). $\square$

9. CLASSIFICATION OVER THE TORUS TATE RING

Proof of Theorem 6.7. Suppose first that k is algebraically closed of characteristic two. The zero algebra is the empty product, so let $B \neq 0$. By Lemma 6.5, $d = \deg(B)$ is finite. Its last nonzero splitting-tower layer $D = B^{[d]}$ satisfies

$$D \otimes B \simeq D^{\times d} \quad \text{as } D\text{-algebras} \quad (9.1)$$

by [Bal14, Theorem 3.9(c)]. Proposition 6.1 lifts the resulting algebra isomorphism in the homotopy category to the displayed equivalence of commutative D -algebras.

The algebra D is separable and perfect in the original category. Indeed, separable algebras in the category of modules over a separable algebra are separable after forgetting to the original category [Bal14, Remark 1.5]. Induction applies this fact to the splitting tower. Relative tensor products over a separable algebra are retracts of ordinary tensor products, and the next tower layer is an algebra factor of such a tensor product [Bal14, §1 and Remark 1.3]. Both preserve perfectness.

Proposition 8.3 applies to the nonzero algebra D . Hence, on every standard chart, D_i is a full-support perfect R_i -algebra. Proposition 6.6(2) makes it descendable, and (9.1) exhibits B_i as a finite cover of R_i . It follows from Proposition 6.6(3) that B_i is the spectral lift of a finite étale $\pi_0 R_i$ -algebra.

These ordinary finite étale algebras glue on the standard overlaps: the spectral lifts and their maps are fully determined by π_0 , so the descent data of B give descent data for the ordinary algebras. They define a finite étale algebra on $\mathbb{P}_k^{r-1}$. Projective space is geometrically simply connected, so its finite étale covers over a separably closed field are constant. This is also the geometric input in [Mat16, proof of Theorem 9.9]. The glued algebra is therefore $\mathcal{O}_{\mathbb{P}_k^{r-1}}^{\times n}$. Unique spectral étale lifting and (6.2) give $B \simeq (k^{t^{\mathbb{T}^r}})^{\times n}$. This includes $r = 1$, where the ordinary scheme is $\mathbb{P}_k^0$.

Now let k be separably closed of characteristic two, without assuming it perfect, and let $\bar{k}$ be its algebraic closure. Base change $\mathfrak{X}_r$ and the perfect algebra B to $\bar{k}$. On a chart,

$$R_i(k) \otimes_{Hk} H\bar{k} \simeq R_i(\bar{k}). \quad (9.2)$$

For cochains, the comparison follows from finite-dimensionality in each cohomological degree; localization then gives the displayed equivalence. The base-changed algebra sheaf remains separable

and perfect. Under (6.2) over $\bar{k}$, the case just proved identifies it with a finite product of copies of the structure sheaf.

We descend étaleness on each chart. Faithful flatness of $\bar{k}/k$ and (9.2) give $\pi_{\text{odd}}B_i = 0$. Put $T_i = \pi_0B_i$. Perfectness makes T_i finite over π_0R_i , and Noetherianity makes it finitely presented. It becomes finite étale after the faithfully flat base change to $\bar{k}$. Consequently, flatness descends, and the module Ω_{T_i/π_0R_i} , which vanishes after this base change, already vanishes over k . Thus T_i is finite étale over π_0R_i .

The periodicity unit in R_i , together with the vanishing of odd homotopy, gives the canonical isomorphism

$$\pi_*R_i \otimes_{\pi_0R_i} T_i \xrightarrow{\sim} \pi_*B_i.$$

Hence B_i is the spectral étale lift of T_i . The same gluing and projective-space argument over k proves the theorem in this case. This descent only uses flat base change on the charts; it does not require global sections to commute with an infinite field extension.

Finally, suppose p is odd. On each standard chart, 2 is invertible and π_0R_i is regular Noetherian. The last assertion of Proposition 6.6(3) applies directly to B_i , making it a spectral finite étale algebra. These algebras glue and are trivialized on projective space exactly as above. $\square$

Remark 9.1. In characteristic two, the proof establishes the finite-cover condition by using the last nonzero tower layer as the descendable splitting algebra. Full support of the original algebra alone would not identify a splitting algebra. Likewise, the commutative residue algebra was constructed only at the rational point forced by power operations; no such construction was assumed at an arbitrary point.

Corollary 9.2 (Whole-group stable triviality). *Let $E = C_p^r$, and let k be separably closed of characteristic p . If $A \in \text{CAlg}^{\text{sep}}(\text{stmod}(kE))$ satisfies $K_A = E$, then $A \simeq k^{\times n}$ for some $n \geq 0$.*

Proof. For $r \geq 1$, Lemma 6.3 gives a separable perfect lift over $k^{t^{\mathbb{T}^r}}$. Apply Theorem 6.7 and forget the torus-fixed structure. If $E = 1$, its stable module category is zero and the assertion holds there as well. $\square$

10. ELEMENTARY ABELIAN GROUPS

Theorem 10.1. *Let k be a separably closed field of characteristic $p > 0$. For every $r \geq 0$, each separable commutative algebra in $\text{stmod}(kC_p^r)$ is standard.*

Proof. Suppose otherwise, and choose a counterexample on $E = C_p^r$ with r minimal. By Lemma 4.1, a nonzero connected factor is again a counterexample; denote it by A . The case $r = 0$ is excluded because the stable category is zero. Lemma 4.3 gives $K_A \neq 1$. If $K_A = E$, Corollary 9.2 makes A standard. We may therefore set $K = K_A$ and choose a complement H such that

$$E = H \times K, \quad H \neq 1, \quad K \neq 1. \quad (10.1)$$

Both factors have rank smaller than r .

Choose a core representative M of A . We will first show that $M \downarrow_H$ is free. Its invariant algebra M^H will then be a nonzero connected core over K . Minimality forces this core to have dimension one, so $M \downarrow_H$ has free rank one. At that rank, the transfer on endomorphisms vanishes, turning the stable triviality of the K -action into actual triviality. Lemma 2.6 will then identify A with an ordinary function algebra.

Restriction to H . By minimality, $A \downarrow_H$ is standard. Suppose it is nonzero. Delete the free orbits of a representing finite H -set and choose a point whose stabilizer is nontrivial. Let $D \leq H$ be a subgroup of order p in that stabilizer. On restriction to D , every orbit is either a fixed point or a free orbit, and there is at least one fixed point. Thus

$$A \downarrow_D \simeq k^{\times m}, \quad m > 0.$$

Every $d \in D$ acts as the identity on this nonzero stable algebra. Indeed, the chosen algebra isomorphism is represented by a D -linear map and hence commutes with the action of d . Restriction is additive and strong symmetric monoidal, so Lemma 4.2 implies that each such d already acts as the identity on A . This gives $D \leq H \cap K = 1$, a contradiction. Hence

$$A \downarrow_H = 0. \quad (10.2)$$

It follows that $M \downarrow_H$ is projective, and thus a nonzero free kH -module by Lemma 2.4.

The dimension of M^H . Proposition 5.1 and Lemma 5.2 make $C = M^H$ a nonzero connected separable algebra in $\text{stmod}(kK)$ whose underlying module is a core. For each $g \in K$, apply the additive functor (5.2) to the stable equality $g|_M = 1_M$. This shows that the action of K on C is stably trivial.

By minimality, C is standard. After free orbits have been discarded, each remaining transitive orbit contributes a nonzero algebra factor. Connectedness therefore leaves exactly one such orbit, giving $C \simeq k(K/L)$ for some $1 \neq L \leq K$. Because K is abelian, L acts actually trivially on this permutation module. Lemma 2.6 implies that its endomorphisms have no projective-factorization ambiguity. The stable triviality of the whole K -action therefore makes the action on $k(K/L)$ actually trivial, so $L = K$ and $C \simeq_{\text{st}} k$. Both modules are cores; Lemma 2.2 now gives an ordinary module isomorphism and hence $\dim_k C = 1$. Equation (5.5) shows that $\dim_k M = |H|$. Since M is free over H , this proves

$$M \downarrow_H \cong kH. \quad (10.3)$$

Actual triviality of the K -action. The ring

$$\text{End}_{kH}(M) \cong (kH)^{\text{op}} = kH$$

is commutative by (10.3) and the commutativity of H . For every k -linear endomorphism f of M , its H -transfer belongs to this ring. The action of each $g \in K$ is also an invertible element of this ring, since it commutes with H . Conjugation by K thus fixes $\text{Tr}_H(f)$, and the iterated transfer formula gives

$$\text{Tr}_E(f) = \sum_{g \in K} g \text{Tr}_H(f) g^{-1} = |K| \text{Tr}_H(f) = 0.$$

The last equality uses $K \neq 1$ and $\text{char}(k) = p$. Lemma 2.5 now gives $\mathcal{P}_E(M, M) = 0$. Therefore $g|_M = 1_M$ holds as an ordinary module-map equality for every $g \in K$. This nontrivial subgroup acts actually trivially on M , so Lemma 2.6 makes the algebra and its separability equations ordinary module-map equations and makes A standard. This contradicts the choice of A . $\square$

11. STANDARD ALGEBRAS AND GROUP ACTIONS

The passage from elementary abelian groups to arbitrary p -groups rests on the following descent statement.

Proposition 11.1 (Descent from maximal subgroups). *Let P be a finite p -group that is not elementary abelian. If a separable commutative algebra $A \in \text{stmod}(kP)$ is standard after restriction to every maximal proper subgroup of P , then A is standard.*

We prove the proposition in Section 12. The first task is to recover finite sets, including their full group actions, from the local standard algebras. This requires a description of maps between standard algebras that does not use the classification theorem.

Lemma 11.2 (Full faithfulness for function algebras). *Let $Q \neq 1$ be a finite p -group, and let X, Y be finite Q -sets with no free Q -orbits. Pullback induces a bijection*

$$\text{Hom}_{Q\text{-Set}}(Y, X) \xrightarrow{\sim} \text{Hom}_{\text{CAlg}(\text{stmod}(kQ))}(kX, kY). \quad (11.1)$$

Proof. Decompose Y into orbits. Since function algebras turn disjoint unions into categorical products of commutative algebras, it suffices to take $Y = Q/K$ with $K \neq 1$. Restriction is strong symmetric monoidal, and its right adjoint, coinduction, carries k to the function algebra $k(Q/K)$. The induced adjunction on commutative algebras gives

$$\mathrm{Hom}_{\mathrm{CAlg}(\mathrm{stmod}(kQ))}(kX, k(Q/K)) \cong \mathrm{Hom}_{\mathrm{CAlg}(\mathrm{stmod}(kK))}(k(X \downarrow_K), k). \quad (11.2)$$

This is the restriction–coinduction description underlying [Bal15, Theorem 1.2]. Explicitly, the adjoint of $\psi : U \downarrow_K \rightarrow k$ is

$$f(u)(gK) = \psi(g^{-1}u).$$

The unit and multiplication identities correspond under this formula. Restriction and coinduction preserve projectives, so the adjunction and these identities descend to the stable categories.

The algebra k is connected: since $K \neq 1$, its stable endomorphism ring is k . Decompose $X \downarrow_K$ into K -orbits. The images of the component units under an algebra map to k are orthogonal scalar idempotents whose sum is 1. Exactly one is 1, and the map is zero on every other component, because that component's unit acts as its identity. Thus the map chooses one orbit factor.

A free orbit contributes the zero stable algebra and cannot be chosen. For $1 < L < K$, a unital map $k(K/L) \rightarrow k$ would split the unit and make k a stable direct summand of a different nonprojective indecomposable. Lemmas 2.3 and 2.7 exclude this. Only a fixed-point factor can therefore be chosen, and its unital map to k is unique. The right side of (11.2) is consequently X^K , identified in the usual way with $\mathrm{Hom}_Q(Q/K, X)$.

The same conclusion covers empty sets. The zero algebra is terminal, and there is no unital map from it to a nonzero algebra. $\square$

Lemma 11.3 (Transport of the full group action). *Let $1 \neq H \leq P$, let $A \in \mathrm{CAlg}(\mathrm{stmod}(kP))$, and choose an isomorphism*

$$A \downarrow_H \simeq kX_H$$

of stable algebras, where X_H is a finite H -set with no free H -orbits. The conjugation maps of A determine a P -action on X_H extending its given H -action.

If $1 \neq K \leq H$ are normal in P and such choices have been made for both subgroups, restriction determines a P -equivariant bijection

$$r_{K,H} : \nu_K(X_H) \xrightarrow{\sim} X_K, \quad \nu_K(X_H) = \{x \in X_H : K \cap P_x \neq 1\}. \quad (11.3)$$

For chains of normal subgroups, these bijections are compatible with composition.

Proof. Represent A by a P -module, with P -linear maps representing its multiplication and unit. These maps satisfy the algebra identities in the stable category; no ordinary algebra structure on this representative is assumed. Write λ_g for the action of $g \in P$ on the module. The P -linearity of the structure maps gives

$$\lambda_g \mu = \mu(\lambda_g \otimes \lambda_g), \quad \lambda_g \eta = \eta.$$

Thus λ_g induces a conjugation-semilinear isomorphism of stable algebras on restriction to H .

Let $\alpha_H : A \downarrow_H \xrightarrow{\sim} kX_H$ be the chosen stable algebra isomorphism. Transporting λ_g through α_H gives a stable semilinear isomorphism t_g of kX_H . Lemma 11.2, applied also to the conjugate H -set, gives a unique bijection $\tau_g : X_H \rightarrow X_H$ with

$$\tau_g(hx) = (ghg^{-1})\tau_g(x).$$

Our convention is that t_g is represented by the permutation of basis vectors

$$e_x \mapsto e_{\tau_g(x)}; \quad \text{equivalently, } t_g = (\tau_g^{-1})^*.$$

Composition of the stable semilinear maps gives $t_g t_{g'} = t_{gg'}$. Full faithfulness then gives $\tau_g \tau_{g'} = \tau_{gg'}$. A representative of α_H is an H -linear module map, so for $h \in H$ it intertwines λ_h with the given action on kX_H . Hence τ_h is the prescribed H -action. This proves the first assertion.

Since K is normal in P , the subset $\nu_K(X_H)$ is P -invariant. Its complement is the union of the free K -orbits. The projection $q_{K,H} : k(X_H \downarrow_K) \rightarrow k\nu_K(X_H)$ is therefore an isomorphism in the stable category over K . The comparison of the chosen algebra isomorphisms is explicitly

$$c_{K,H} = \alpha_K(q_{K,H} \alpha_H \downarrow_K)^{-1} : k\nu_K(X_H) \xrightarrow{\sim} kX_K.$$

Define $r_{K,H}$ by the convention that $c_{K,H}$ sends e_x to $e_{r_{K,H}(x)}$. Full faithfulness gives its existence and uniqueness. The comparison commutes with all the transported semilinear maps, so $r_{K,H}$ is P -equivariant.

If $1 \neq L \leq K \leq H$ are normal in P , transitivity of restriction and cancellation of the intermediate chosen isomorphism give

$$r_{L,K} \circ \nu_L(r_{K,H}) = r_{L,H}. \quad (11.4)$$

The domains agree because $\nu_L \nu_K(X_H) = \nu_L(X_H)$: a nontrivial L -stabilizer is also a nontrivial K -stabilizer. $\square$

Remark 11.4. The action supplied by Lemma 11.3 is an action of P , not an action of P/H on the underlying set. For representatives of cosets in P/H , the discrepancy between their product and the chosen representative of that product lies in H and acts by the original H -action. The relations $\tau_g \tau_{g'} = \tau_{gg'}$ retain this information even when the extension $1 \rightarrow H \rightarrow P \rightarrow P/H \rightarrow 1$ does not split.

12. GLUING OVER NORMAL MAXIMAL SUBGROUPS

We use subgroup descent in the following form. Let $\mathcal{F}$ be a collection of subgroups of P , closed under conjugation and nonempty finite intersections. If every elementary abelian p -subgroup is contained in a member of $\mathcal{F}$, then

$$\mathrm{StMod}(kP) \simeq \varprojlim_{P/H \in \mathcal{O}_{\mathcal{F}}(P)^{\mathrm{op}}} \mathrm{StMod}(kH) \quad (12.1)$$

as symmetric monoidal stable ∞ -categories. Here $\mathcal{O}_{\mathcal{F}}(P)$ is the full orbit category on the objects P/H with $H \in \mathcal{F}$; its diagram retains restriction, conjugation, and their coherences. We use the argument for [Mat16, Corollary 9.16] over an arbitrary field of characteristic p , although Section 9 of that reference assumes the field perfect. Put $B = \prod_{H \in \mathcal{F}} k(P/H)$. The elementary abelian descent theorem holds over discrete coefficient rings [Mat16, Theorem 4.8]. Since the restriction of B to every elementary abelian subgroup contains the unit as a direct factor, the proof of [Mat16, Proposition 9.13] makes B descendable in $\mathrm{Fun}(BP, \mathrm{Mod}(k))$. The finite cobar construction witnessing descendability lies among k -perfect representations. Applying the exact symmetric monoidal stable quotient to this finite construction makes the image of B descendable in $\mathrm{StMod}(kP)$. The module-category identification of [Mat16, Proposition 9.12] and the coefficient-independent orbit-category cofinality argument then give (12.1): stabilizers in nonempty products of the orbits P/H are intersections of conjugates, hence remain in $\mathcal{F}$.

Proof of Proposition 11.1. We first choose a family of nontrivial normal subgroups to which (12.1) applies. The local finite sets are then glued one P -orbit type at a time. Finally, their equivariance identifies the conjugation data required for algebra descent.

The family of intersections. Every maximal subgroup of a finite p -group is normal of index p . Indeed, every proper subgroup $H < P$ is strictly contained in its normalizer. To prove this by induction, adjoin a central element outside H if $Z(P) \not\subseteq H$; otherwise apply induction to $H/Z(P) < P/Z(P)$. A maximal subgroup is therefore normal, and the quotient has no nontrivial proper subgroup, so its order is p .

Let $M_1, \dots, M_s$ be the maximal subgroups and put $\Phi(P) = \bigcap_i M_i$. Each P/M_i is abelian of exponent p , so all commutators and all p -th powers belong to $\Phi(P)$. If $\Phi(P)$ were trivial, P would

be elementary abelian. Consequently

$$\Phi(P) \neq 1. \quad (12.2)$$

Take $\mathcal{F}$ to be the intersections of nonempty subsets of $\{M_1, \dots, M_s\}$. Its members are proper normal subgroups containing $\Phi(P)$, and the family is closed under intersections and conjugation. Every elementary abelian subgroup is proper in P and hence lies in a maximal subgroup. Thus subgroup descent applies, and every subgroup in the indexing family is nontrivial.

For $H \in \mathcal{F}$, restriction from a maximal subgroup containing H shows that $A \downarrow_H$ is standard. Choose $A \downarrow_H \simeq kX_H$ and delete the free H -orbits. Lemma 11.3 makes each X_H a finite P -set and provides the equivariant transitions $r_{K,H}$ satisfying (11.4).

A common model for each orbit type. For a conjugacy class $[L]$ of subgroups of P , let $X_H^{(L)}$ be the union of the P -orbits in X_H isomorphic to P/L . Only $L \neq 1$ occurs, since X_H has no free H -orbits. Normality of H gives

$$H \cap gLg^{-1} = g(H \cap L)g^{-1}.$$

Thus an entire orbit P/L survives deletion of the free H -orbits exactly when $H \cap L \neq 1$. In particular, for $K \leq H$ the map $r_{K,H}$ restricts to a bijection of the (L) -parts whenever $K \cap L \neq 1$.

For $1 \neq L < P$, choose a maximal subgroup M_L containing L ; for $L = P$, choose any maximal subgroup M_L . Set

$$Y_L = X_{M_L}^{(L)}.$$

Suppose $H \cap L \neq 1$, and put $J = H \cap M_L$. Then $J \cap L \neq 1$: for $L < P$ this follows from $L \leq M_L$, while for $L = P$ it follows from $\Phi(P) \leq J$ and (12.2). Both transitions in

$$Y_L \xrightarrow{r_{J,M_L}} X_J^{(L)} \xrightarrow{r_{J,H}^{-1}} X_H^{(L)} \quad (12.3)$$

are therefore bijections. This comparison only uses overlaps on which the orbit type has not been deleted.

These identifications commute with restriction. To see this for $K \leq H$ with $K \cap L \neq 1$, put $J_K = K \cap M_L$ and $J_H = H \cap M_L$. The two comparisons become equal after composition with $r_{J_K,K}$, by the cocycle identities for $J_K \leq J_H \leq M_L$, $J_K \leq J_H \leq H$, and $J_K \leq K \leq H$. Since $r_{J_K,K}$ is a bijection on this orbit type, the original comparisons agree. If $K \cap L = 1$, both restrictions of this orbit type are deleted.

Define

$$X = \coprod_{[1 \neq L \leq P]} Y_L. \quad (12.4)$$

This is a finite P -set with no free P -orbits. The preceding comparisons give P -equivariant bijections

$$\theta_H : \nu_H(X) \xrightarrow{\sim} X_H \quad (H \in \mathcal{F}) \quad (12.5)$$

such that $\theta_K = r_{K,H} \circ \nu_K(\theta_H)$ for $K \leq H$. It remains to check that these bijections identify the full orbit category diagrams, rather than only their restriction maps.

The orbit category comparison. For normal $K, H \in \mathcal{F}$, a map $P/K \rightarrow P/H$ exists exactly when $K \leq H$. In that case every map has the form

$$f_g : P/K \longrightarrow P/H, \quad xK \longmapsto xgH, \quad gH \in P/H.$$

It is the inclusion projection followed by an automorphism of the target orbit. On modules, the associated functor sends an H -module V to its underlying vector space with K -action $k \cdot v = (g^{-1}kg)v$. For a restricted P -module, its canonical comparison with the ordinary K -restriction is multiplication by g . Replacing g by gh , with $h \in H$, identifies the two twisted restrictions by multiplication by h ; their comparison maps agree because $\lambda_{gh} = \lambda_g \lambda_h$. For $f_t : P/J \rightarrow P/K$ and $f_g : P/K \rightarrow P/H$, with $J \leq K \leq H$ normal in P , one has $f_g \circ f_t = f_{tg}$. The contravariant module comparisons compose as $\lambda_t \lambda_g = \lambda_{tg}$. The set-theoretic counterparts are precisely the transported maps τ_g and the relation $\tau_{gh} = \tau_g \tau_h$ of Remark 11.4.

The bijections (12.5) commute with these comparisons: for $x \in \nu_K(X)$,

$$\theta_K(gx) = r_{K,H}(g\theta_H(x)).$$

Indeed, this follows from their P -equivariance and compatibility with $r_{K,H}$. Thus they identify the arrows f_g for A and kX , as well as their composition and representative relations.

Proposition 6.1 identifies separable commutative algebras with their homotopy-category counterparts, gives discrete mapping spaces, and states that their category commutes with limits of stable symmetric monoidal categories and exact symmetric monoidal functors. The preceding arrow comparisons consequently identify the full algebra descent data: there are no further higher homotopies to choose. Applying (12.1) gives $A \simeq kX$. Both algebras are finite stable modules, so this is an isomorphism in $\text{stmod}(kP)$. $\square$

Proof of Theorem 1.1. We prove essential surjectivity by induction on $|P|$. For $P = 1$, the stable category is zero and the only finite P -set without free orbits is the empty set. For nontrivial elementary abelian groups, the assertion is Theorem 10.1. Otherwise every maximal subgroup has smaller order, and restriction preserves separability and commutativity. The induction hypothesis makes the restrictions standard, so Proposition 11.1 makes A standard.

Free orbits may be deleted: their function algebras are projective, and projection onto the function algebra of the nonfree part is a stable algebra isomorphism. Full faithfulness is Lemma 11.2. Together these observations prove (1.2). Every finite P -set without free orbits has a unique decomposition into transitive sets P/H , with $H \neq 1$ taken up to conjugacy. The contravariant function-algebra functor therefore gives (1.3), including the uniqueness of the multiplicities. $\square$

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